Instruction Routing in Vision-Language-Action Models: A Cross-Architecture Token-Level Analysis

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Abstract

Vision-Language-Action models (VLAs) fine-tune large language models on robot demonstration data, yet how training reshapes instruction routing and information flow at the token level remains uncharacterized—and whether such effects generalize across architectures is unknown. We present a systematic experimental framework for token-level analysis of decoder-only VLAs, validated across three architectures (OpenVLA/Llama-2-7B, TracVLA/Phi-3-mini, NORA/Qwen2.5-VL-3B) and four LIBERO task suites. We report three main findings. First, VLA fine-tuning reorganizes instruction attention from the mid-layer processing pattern of pretrained LLMs to an entry-exit bookend pattern, achieving immediate instruction routing at layer 1 (99.6% classification accuracy vs. 80.2% for the pretrained LLM baseline). Second, R^2 exit collapse—a sharp drop in action-decodability at exit layers—consistently emerges when VLMs are first fine-tuned into VLAs (6/6 cells across two architectures, $\Delta = 0.138$–0.691) but not during subsequent VLA-to-VLA fine-tuning (0/4 cells, $\Delta < 0.086$), indicating a one-time representational reorganization. Third, under RSA, vision encoder geometry follows an inverted-U profile with peak alignment at mid-layers, invariant to both training state and instruction content ($\Delta\rho < 0.017$). An NLP fine-tuning control ($\Delta R^2 = 0.000$) confirms these effects are specific to action-oriented training. These results characterize a three-tier routing gradient—from VLM base through general VLA to task-specific VLA—suggesting that instruction routing is a staged emergent property of VLA training.

1 Introduction

Vision-Language-Action models (VLAs) adapt large language models to robotic control by fine-tuning on paired observation-action data, conditioned on natural language instructions [Brohan *et al.*, 2023, Kim *et al.*, 2024]. Despite strong task performance, the internal mechanisms by which these models integrate visual observations with language instructions remain poorly understood. Prior behavioral studies have documented visual dominance and instruction underutilization in VLA policies [Grant *et al.*, 2026, Haon *et al.*, 2025], but these analyses operate at the input–output level and do not decompose the representational roles of different token positions or explain how training reshapes information flow.

We address this gap with a systematic experimental framework for token-level analysis of decoder-only VLAs. Our framework combines six complementary analysis methods—Representational Similarity Analysis (RSA), Centered Kernel Alignment (CKA), linear and MLP action probes (R^2), cross-instruction classification, attention profiling, and counterfactual perturbation—applied in a factorial design across training conditions (trained, pretrained base, untrained) and instruction configurations. We validate this framework across three VLA architectures built on distinct LLM backbones: OpenVLA (Llama-2-7B), TracVLA (Phi-3-mini), and NORA (Qwen2.5-VL-3B), tested on four LIBERO task suites (Goal, Spatial, Object, Long), yielding a 3×4 evaluation matrix with 10 filled cells.

Our analysis reveals three principal findings:

(1) Immediate dedicated routing (C2). VLA fine-tuning reorganizes instruction processing from the mid-layer inverted-U pattern characteristic of pretrained LLMs (peak at layers 9–16) to an entry-exit bookend pattern (peaks at layers 0–1 and 31). Three qualitatively different routing strategies emerge: trained VLAs achieve immediate instruction capture at layer 1 (99.6%), pretrained LLMs require gradual buildup (80.2% at L1, saturating at L5), and random-weight models exhibit accidental routing that decays (100% at L1 declining to 82% at L31).

(2) Cross-architecture R^2 exit collapse. Action-decodability collapses at exit layers when VLMs are first converted to VLAs (6/6 cells, $\Delta = 0.138$–0.691) but not during further VLA-to-VLA fine-tuning (0/4 cells, $\Delta < 0.086$). An NLP fine-tuning control ($\Delta R^2 = 0.000$) excludes generic fine-tuning as an explanation. This binary VLM→VLA vs. VLA→VLA separation constitutes a mechanistic signature of initial action training.

(3) Vision encoder geometry (observation, under RSA). The frozen vision encoder provides the dominant source of action-aligned geometric structure (RSA $\rho = 0.615$), which is invariant to training state and instruction content ($\Delta\rho < 0.017$ across a 2×2 factorial). Under RSA, DINOv2 achieves the highest align-

ment ($\rho = 0.692$), though CKA reverses this ordering, highlighting metric dependence.

These findings are unified by a three-tier routing gradient: VLM base (no action routing) → OXE-pretrained VLA (general routing) → task-finetuned VLA (task-specific routing with exit collapse).

2 Related Work

Vision-Language-Action Models. RT-2 [Brohan *et al.*, 2023] demonstrated that VLMs can be adapted for robotic control by tokenizing actions as text. OpenVLA [Kim *et al.*, 2024] scaled this approach with open-source training on the Open X-Embodiment dataset [Open X-Embodiment, 2023]. TracVLA and NORA represent recent architectural variants using Phi-3-mini and Qwen2.5-VL backbones, respectively.

Behavioral Analysis of VLAs. Grant *et al.* [Grant *et al.*, 2026] and Haon *et al.* [Haon *et al.*, 2025] characterized visual dominance and instruction underutilization through perturbation studies. Our work extends this line by moving from behavioral to representational analysis, decomposing token-level information flow.

Mechanistic Interpretability of LLMs. Probing [Alain & Bengio, 2017, Hewitt & Liang, 2019], attention analysis [Clark *et al.*, 2019], and representational similarity [Kornblith *et al.*, 2019, Kriegeskorte *et al.*, 2008] have been widely applied to language and vision-language models. We adapt these tools to the VLA setting, where the additional action modality creates novel analytical challenges.

3 Experimental Framework

3.1 Models and Benchmarks

We analyze three VLA architectures spanning distinct LLM backbones and attention configurations:

- **OpenVLA**: SigLIP + DinoSigLIP vision encoder, Llama-2-7B backbone (32 layers, 32 attention heads). Pretrained on 970K Open X-Embodiment episodes; our "untrained" baseline is the OXE-pretrained checkpoint before LIBERO fine-tuning.
- **TracVLA**: Phi-3-mini backbone (32 layers, 32 heads). The "untrained" baseline is the Phi-3-Vision VLM before any action training.
- **NORA**: Qwen2.5-VL-3B backbone (36 layers, 16 heads with GQA). The "untrained" baseline is the Qwen2.5-VL model.

All models are evaluated on LIBERO [Liu *et al.*, 2024] task suites: Goal (10 tasks), Spatial (10 tasks), Object (10 tasks), and Long (10 tasks). OpenVLA is tested on all four suites; TracVLA on all four; NORA on Goal and Spatial (no Object/Long checkpoints available).

Table 1: Weight change distribution by component between trained and untrained OpenVLA checkpoints.

Component	#Params	Rel. change	cos_sim	Verdict
Vision backbone	685	0.000	1.000	Identical
Projector	6	0.000	1.001	Identical
LLM (32 layers)	288	1.316	0.218	Distributed
LM head	1	1.475	0.000	Changed

3.2 Frozen Encoder Verification

We verify architectural assumptions by comparing weight tensors between trained and untrained checkpoints (Table 1).

The vision backbone and projector are identical across conditions (cos_sim = 1.0), confirming the frozen encoder design. LLM weight changes are uniformly distributed across all 32 layers (rel. change range 1.22–1.37), indicating distributed rather than localized adaptation. The head-to-LLM ratio of 1.12 confirms that action mapping is not a shallow output adaptation.

3.3 Analysis Methods

Our framework employs six complementary methods:

Representational Similarity Analysis (RSA) measures geometric alignment between neural representations and action-distance structure using Spearman correlation over pairwise distance matrices.

Centered Kernel Alignment (CKA) captures global representational structure invariant to orthogonal transformations, complementing RSA's geometric sensitivity.

MLP Action Probes (R^2) train per-layer regressors to predict continuous actions, measuring functional action-decodability at each layer.

Cross-Instruction Classification uses the same visual frame under 10 different task instructions for 10-way classification, isolating instruction routing from visual processing.

Attention Profiling quantifies instruction-dominant heads (>50% attention to text tokens after per-token normalization) across all layers.

Counterfactual Perturbation measures action sensitivity to instruction modifications (empty, shuffled, wrong, random tokens) via Cohen's d or cosine distance.

4 Results: Vision Encoder Geometry

4.1 Action-Aligned Geometry: Source Localization

We first localize the source of action-relevant geometric structure by comparing RSA across extraction points

Table 2: Action-aligned geometry: source localization via RSA.

Representation	RSA (ρ)	95% CI	vs DINO-A
SigLIP+Projector (4096d)	0.615	[0.526, 0.695]	+0.227
VLA Trained peak (L9)	0.488	[0.41, 0.56]	+0.100
VLA Untrained peak (L12)	0.485	[0.40, 0.56]	+0.097
DINOv2 cls_token	0.388	[0.30, 0.47]	—

Table 3: DINOv2 standalone baselines.

Feature	RSA (ρ)	Action R^2	Temporal ρ
DINOv2 cls_token	0.692	0.687	0.901
DINOv2 patch_mean	0.631	—	—
Label-permuted control	—	−0.052	0.039

Table 4: Cross-encoder comparison (unified probe).

Encoder	RSA (ρ)	Action R^2	Pretraining
DINOv2 cls	0.692	0.634	Self-supervised
SigLIP+Proj	0.615	0.630	VL contrastive
CLIP ViT-L cls	0.557	0.587	VL contrastive
VLA last_pre L19	—	0.671	VLA finetuned
VLA img_mean L19	—	0.599	VLA finetuned

Table 5: Branch-split RSA analysis within fused backbone.

Representation	Dim	RSA (ρ)	Δ vs Fused
DINOv2 branch	1088	0.619	−0.015
SigLIP branch	1088	0.621	−0.013
Fused backbone	2176	0.634	—
Post-projector	4096	0.615	−0.019
Standalone DINOv2 cls	1024	0.692	+0.058

(Table 2).

The frozen vision encoder (SigLIP+Projector) provides RSA $\rho = 0.615$, substantially higher than any VLA layer ($\sim$0.49) or standalone DINOv2 (0.388 within the VLA pipeline). The LLM degradation of $\Delta = -0.127$ reflects the cost of integrating visual features with language representations—a functional trade-off rather than information loss.

4.2 DINOv2 Baseline and Cross-Encoder Comparison

As a standalone encoder, DINOv2 ViT-L/14 achieves the highest RSA among all tested encoders (Table 3).

The label-permutation control ($\rho = 0.039 \approx 0$) excludes probe overfitting. Cross-encoder comparison (Table 4) reveals that DINOv2's self-supervised pretraining produces higher action-aligned geometry than contrastive VL pretraining (CLIP RSA = 0.557, R^2 = 0.587), suggesting spatial feature quality matters more than language grounding for action prediction.

4.3 Branch-Split Analysis

Within OpenVLA's fused DinoSigLIP backbone, we isolate each branch's contribution (Table 5).

The two branches contribute nearly equally ($\Delta < 0.002$), and fusion provides a complementary benefit ($0.634 > 0.619, 0.621$). The projector introduces mild compression ($0.634 \rightarrow 0.615$), acting as an information bottleneck that preserves core geometric structure.

4.4 Per-Layer RSA Profile

The per-layer RSA curve reveals qualitatively different patterns for trained and untrained models (Table 6).

The trained model exhibits an inverted-U profile (peak at L12, $\rho = 0.628$) while the untrained model decays monotonically from L0. Training shifts the RSA peak from L0 to L12, temporarily preserving action geometry at mid-layers before exit-layer collapse ($0.628 \rightarrow 0.422$).

4.5 CKA Verification and Metric Divergence

CKA analysis reveals an ordering reversal relative to RSA (Table 7), an important methodological finding.

RSA captures geometric alignment (action-distance correspondence) while CKA captures global structure (rotation-invariant similarity). The LLM degrades the former but preserves the latter. DINOv2's top ranking under RSA does not hold under CKA, motivating our qualification of C3 as "under RSA." This dissociation is itself informative: it reveals that LLM processing disrupts fine-grained geometric alignment while maintaining broader representational structure.

4.6 Six-Method Consistency

Six independent analysis methods converge on a consistent pattern (Table 8): trained $\approx$ untrained $\gg$ DINO across all metrics.

This multi-method convergence validates the framework's robustness: the frozen encoder dominates representational quality regardless of the metric chosen.

4.7 Instruction Invariance: Factorial Design

A 2×2 factorial design (training $\times$ instruction) confirms that image-token geometry is doubly invariant (Table 9).

Instruction removal produces $\Delta = -0.004$ (Table 10), far below the 0.10 threshold, consistent with the causal mask preventing information flow from instruction tokens to preceding image tokens.

Instruction content ablation (Table 11) further confirms that neither instruction presence, content,

Table 6: Per-layer RSA profile (image_mean, with instruction).

Layer	Trained RSA	Untrained RSA	Δ(T−U)
Projector	0.615	0.615	0.000
L0	0.612	0.612	0.000
L4	0.472	0.470	+0.002
L8	0.512	0.478	+0.034
L12	0.628	0.476	+0.152
L16	0.523	0.470	+0.053
L24	0.476	0.460	+0.016
L32	0.422	0.545	−0.123

Table 7: RSA vs. CKA ordering comparison.

Rank	RSA Ordering	Linear CKA Ordering
1st	SigLIP+Proj (0.615)	last_pre L32 (0.127)
2nd	Trained img (0.488)	Untrained img (0.089)
3rd	Untrained img (0.485)	Trained img (0.083)
4th	DINO (0.388)	Projector (0.081)
5th	—	DINO (0.066)

nor semantic structure affects image-token geometry ($\max|\Delta| = 0.017$). Together with the factorial design, this establishes three-way invariance: training + instruction presence + instruction content all fail to modulate image_mean RSA.

5 Results: Instruction Routing

5.1 Three Qualitatively Different Routing Strategies

Cross-instruction classification across 33 layers reveals three qualitatively distinct routing strategies (Table 12).

Trained VLA: Immediate routing—instruction identity is fully captured at L1 (99.6%) and maintained through all subsequent layers (100%). **Pretrained LLM (Llama-2 base)**: Gradual buildup—L1 achieves 80.2%, saturating at L5. Once resolved, instruction signal is stably maintained. **Untrained (random init)**: Accidental routing—high-dimensional random features produce trivial separability at L1 (100%), but noise accumulation causes signal decay to 82% at L31.

LOOCV confirms: trained achieves $100.00 \pm 0.00\%$ (L8–L31), while untrained decays from $96.64 \pm 7.07\%$ to $84.76 \pm 11.59\%$. Image_mean remains at chance (10%) across all conditions and layers, confirming the causal mask prevents image → instruction information flow.

5.2 L1 Attention: Three-Condition Gradient

The mechanistic basis for the routing differences is revealed in L1 attention patterns (Table 13).

Table 8: Six-method consistency summary.

Method	Trained	Untrained	DINO	T≈U?
Temporal ρ	0.955	0.933	0.901	Yes
Action R^2	0.683	0.616	0.676	Marginal
Delta R^2	0.059	0.049	0.060	Yes
RSA (ρ)	0.488	0.485	0.388	Yes
CKA	0.083	0.089	0.066	Yes
Classification	1.000	1.000	0.863	Yes

Table 9: 2×2 factorial RSA matrix (image_mean, peak layer). Bootstrap CI = [−0.058, 0.072] includes 0.

	With Inst.	No Inst.	Inst. Effect
Trained	0.488	0.489	−0.001
Untrained	0.485	0.480	+0.004
Training Effect	+0.003	+0.009	

VLA training amplifies per-token text/image attention ratio from 1.5× (untrained) to 38.2× (trained), creating 24/32 instruction-dominant heads. This 25.5× amplification reflects genuine instruction specialization, not a token-count artifact (per-token normalization accounts for the ∼11.5× image/text token count asymmetry).

5.3 32-Layer Dense Attention Profile

The full 32-layer attention profile reveals a fundamental reorganization of instruction processing (Table 14).

Untrained: Zero instruction-dominant heads at all layers—no language processing capacity. **Llama-2 base**: Inverted-U (L9–L16 peak at 15–22 heads)—the classical LM pattern of mid-layer deep semantic processing. **VLA Trained**: Bookend (L0–L1 peak at 21–24 heads, L31 re-engagement at 18 heads)—entry-layer instruction capture followed by vision-dominated mid-layer computation, then exit-layer instruction re-injection for action conditioning.

This temporal reorganization from mid-peak to bookend is the mechanistic signature of VLA fine-tuning's routing contribution.

5.4 Full-Layer R^2 Scan

MLP action probes across 33 layers × 2 conditions × 2 token aggregations provide functional validation (Table 15).

Trained last_preaction achieves $R^2 = 0.678$ (L13), exceeding the projector baseline (0.630, $p < 0.001$), confirming that the LLM enhances action-decodability at instruction-conditioned positions. The key dissociation: training effect is small for image_mean ($\Delta = 0.015$, $p = 0.006$) but large for last_preaction ($\Delta = 0.069$, $p < 0.0001$), confirming that training reshapes instruction routing rather than vision geometry. Untrained

Table 10: Instruction ablation (untrained, image_mean).

Condition	Peak RSA	Peak Layer	Δ vs with-inst
With instruction	0.485	L12	—
No instruction	0.480	L6	−0.004
Gate threshold	—	—	0.10

Table 11: Instruction content ablation (trained, image_mean).

Instruction Type	Peak RSA	Peak Layer	Δ vs Normal
Normal (real task)	0.489	L16	—
Random tokens	0.484	L16	−0.005
Word-shuffled	0.483	L16	−0.006
Empty string	0.472	L16	−0.017

models exhibit pathological $R^2 \approx 0$ at layers L10–L13, contrasting sharply with the trained model's smooth profile.

5.5 Counterfactual Action Analysis

Head ablation at L1 quantifies the functional impact of instruction routing on action output (Table 16).

Ablating instruction-dominant heads produces $d = 0.651$ action shift without systematic accuracy degradation (ΔMSE vs. GT ≈ 0, $p = 0.519$). This information-computation dissociation indicates that instruction heads modulate output direction (which action space region) without degrading action quality (distance from ground truth), consistent with the distributed routing hypothesis.

5.6 Training Trajectory

A four-condition developmental trajectory traces routing emergence (Table 17).

Entry routing emerges during OXE pretraining ($0 \rightarrow 24$ heads) and stabilizes during task fine-tuning ($24 \rightarrow 22$). Exit engagement evolves from bookend (18 heads at L31 in OXE-pretrained) to entry-only (0 heads after LIBERO fine-tuning). Critically, probe accuracy remains at 100% at L31 despite zero attention heads—an attention-representation dissociation showing that the residual stream maintains instruction information even when attention is redirected.

6 Results: Cross-Architecture Validation

6.1 R^2 Exit Collapse Matrix

The 3×4 exit collapse matrix reveals a clean binary separation between VLM→VLA and VLA→VLA transitions (Table 18).

Table 12: Cross-instruction classification: three routing strategies (last_preaction token, OpenVLA). Image_mean = 10% (chance) at all layers for all conditions.

Layer	VLA Trained	Llama-2 Base	Untrained
L0	10%	10%	10%
L1	99.6%	80.2%	100%
L2	100%	98.7%	100%
L5	100%	100%	99.x%
L8	100%	100%	98.6%
L16	100%	100%	92.3%
L24	100%	100%	86.4%
L31	100%	100%	82.0%

Table 13: L1 attention pattern: three-condition gradient (OpenVLA).

Metric	Untrained	Llama-2 Base	VLA Trained
Attn to text/instr	11.7%	36.5%	57.7%
Attn to image	87.8%	61.7%	21.3%
Instr-dom heads	0/32	9/32	24/32
Per-token ratio	1.5×	—	38.2×

R^2 exit collapse consistently emerges when VLMs are first fine-tuned into VLAs (6/6 cells, $\Delta = 0.138$–0.691). When an already-pretrained VLA undergoes further task-specific fine-tuning, no cell exceeds the $\Delta > 0.10$ threshold (0/4 OpenVLA cells, $\Delta = 0.004$–0.037). This indicates exit collapse is a one-time representational reorganization associated with the VLM-to-VLA transition rather than a cumulative effect of continued training. NORA-Goal exhibits the most extreme collapse ($\Delta = 0.691$), possibly reflecting the smaller model capacity of Qwen2.5-VL-3B.

6.2 NLP Fine-tuning Control

To exclude generic fine-tuning as an explanation, we analyze three NLP-finetuned models sharing the same backbones (Table 19).

All three NLP-finetuned models exhibit flat $R^2 \approx 0.155$ (noise floor) across all layers with $\Delta R^2 = 0.000$. This dual exclusion—same backbone with different fine-tuning (NLP vs. VLA) and different backbones with same fine-tuning type (NLP instruction tuning)—rules out both backbone-specific and generic-fine-tuning explanations for exit collapse. A chat-finetuned control sharing the same Llama-2-7B backbone confirms that exit collapse is specific to action-oriented training.

6.3 Cross-Architecture Counterfactual Sensitivity

Counterfactual sensitivity to instruction perturbation is consistently present across architectures (Table 20).

Table 14: Dense attention profile: instruction-dominant heads per layer.

Layer	Untrained	Llama-2 Base	VLA Trained	Pattern
L0	0	2	21	Entry peak
L1	0	4	24	Entry peak
L4	0	8	5	Valley
L9	0	18	3	Valley
L12	0	22	1	Valley
L16	0	20	4	Valley
L24	0	12	7	Valley
L31	0	8	18	Exit re-engage

Table 15: R^2 per-layer peak summary (OpenVLA, LIBERO-Goal).

Cond.	Token	Peak L	Peak R^2	Exit R^2	Δ
Trained	img_mean	L8	0.655	0.422	0.233
Trained	last_pre	L13	0.678	0.609	0.069
Untrained	img_mean	L0	0.640	0.545	0.095
Untrained	last_pre	L32	0.609	0.609	0.000
Projector	—	—	0.630	—	—

OpenVLA is the only architecture permitting fair trained-vs-untrained comparison (both conditions generate actions, enabling the same Cohen's d metric). Trained models consistently show higher sensitivity (Δ = 0.134–0.237 for Goal and Spatial). The "empty" condition (complete instruction removal) produces the largest perturbation in 6/8 trained settings, consistent with instruction routing's functional role. We note that 5/8 cross-architecture cells involve incommensurable metrics (Cohen's d for trained, cosine distance for untrained), precluding quantitative comparison.

6.4 Three-Tier Routing Gradient

Integrating exit collapse, counterfactual sensitivity, and L1 attention across training stages reveals a three-tier routing gradient (Table 22).

This gradient reveals routing as a staged emergent property: (1) VLM bases exhibit no action routing; (2) OXE pretraining creates general routing with strong counterfactual sensitivity (d = 1.048) but no exit collapse; (3) task-specific fine-tuning triggers exit collapse and further sharpens counterfactual sensitivity. The NLP control confirms that language-only fine-tuning does not produce any of these effects.

6.5 L1 Heads: Training-Induced vs. Base-Model-Dependent

Cross-architecture L1 analysis distinguishes inherited from training-induced attention patterns (Table 23).

L1 instruction-dominant head counts are base-model-dependent, not training-induced. TracVLA's untrained baseline (pure Phi-3-Vision) already exhibits 32/32 instruction-dominant heads; training reduces this to 27–30/32 as some heads are repurposed for mixed-modality processing. NORA shows minimal change (13/16 → 12–13/16). This counter-intuitive finding—training does not increase L1 head count—distinguishes what training creates (exit collapse, counterfactual sensitivity) from what it inherits (L1 attention patterns from base model pretraining).

Table 16: Counterfactual action analysis (OpenVLA): ablating 24 instruction-dominant vs. 8 control heads at L1.

Metric	Instr. 24h	Control 8h	Ratio
Cohen's d (token)	0.651	0.410	1.59
Trajectory d	2.943	—	—
Paired MSE	0.042	0.020	2.07
Token divergence	0.723	0.293	2.47
KL divergence	5.079	1.690	3.00
GT MSE delta	−0.001 (n.s.)	+0.008 (n.s.)	—

Table 17: Training trajectory: 4-condition L1 evolution (OpenVLA).

Stage	L1 Hd	L31 Hd	L31 Probe	Interp.
Random init	0/32	0/32	82%	No routing
Llama-2 base	9/32	8/32	100%	Language proc.
OXE-pretrained	24/32	18/32	100%	Bookend
LIBERO-finetuned	22/32	0/32	100%	Entry-only

6.6 Cross-Benchmark Consistency

LIBERO-Spatial replicates all core patterns observed in LIBERO-Goal (Table 24).

All core patterns replicate within a 2-layer tolerance ($p < 0.001$ for all trained vs. untrained comparisons), confirming that the representational phenomena are not task-specific artifacts.

6.7 TracVLA Universal Mixed Heads

In TracVLA, heads h13 and h22 consistently transition from instruction-dominant to mixed function across all four LIBERO suites after training (Table 25).

These universal mixed heads may perform cross-modal integration, simultaneously processing image and instruction information. Their cross-suite consistency (4/4) rules out task-specific artifacts, though we note this pattern is TracVLA-specific and may reflect Phi-3-mini's pretraining structure.

7 Discussion

Routing as staged emergence. Our three-tier gradient—VLM base → general VLA → task-specific

Table 18: R^2 exit collapse: VLM→VLA vs. VLA→VLA binary separation. Collapse defined as trained $\Delta > 0.10$.

Architecture	Backbone	Suite	Transition	Trained Δ	Untrained Δ	Collapse?
TracVLA	Phi-3-mini	Goal	VLM→VLA	0.228	0.048	Yes
TracVLA	Phi-3-mini	Spatial	VLM→VLA	0.138	0.016	Yes
TracVLA	Phi-3-mini	Object	VLM→VLA	0.294	0.009	Yes
TracVLA	Phi-3-mini	Long	VLM→VLA	0.198	0.028	Yes
NORA	Qwen2.5-VL-3B	Goal	VLM→VLA	0.691	0.044	Yes
NORA	Qwen2.5-VL-3B	Spatial	VLM→VLA	0.302	0.037	Yes
OpenVLA	Llama-2-7B	Goal	VLA→VLA	0.016	0.033	No
OpenVLA	Llama-2-7B	Spatial	VLA→VLA	0.037	0.034	No
OpenVLA	Llama-2-7B	Object	VLA→VLA	0.037	0.076	No
OpenVLA	Llama-2-7B	Long (OFT)	VLA→VLA	0.004	0.016	No
NLP Control	Llama-2-7B	—	LLM→Chat	0.000	—	No

Table 19: NLP fine-tuning control: R^2 exit collapse comparison.

Model	Finetuning	Peak R^2	ΔR^2
Llama-2-7b-chat	SFT+RLHF	0.1555	0.000
Phi-3-instruct	Instr. tuning	0.1555	0.000
Qwen2.5-3B-Inst	Instr. tuning	0.1555	0.000
TracVLA (trained)	VLM→VLA	0.839	0.265
NORA (trained)	VLM→VLA	0.697	0.701
OpenVLA (trained)	VLA→VLA	0.869	0.016

Table 20: Counterfactual sensitivity: OpenVLA fair comparison (same metric, Cohen's d).

Suite	Trained d	Untrained d	Δ	Top cond.
Goal	1.182	1.048	+0.134	wrong
Spatial	1.173	0.936	+0.237	empty
Object	1.210	cos 0.569*	incomm.	empty
Long	0.565	—	—	shuffled

Table 21: Counterfactual sensitivity: cross-architecture (trained only).

Arch	Suite	Top d/cos	Cond.	Metric
OpenVLA	Goal	d=1.182	wrong	Cohen's d
OpenVLA	Spatial	d=1.173	empty	Cohen's d
TracVLA	Goal	d=1.524	empty	Cohen's d
TracVLA	Spatial	d=1.327	empty	Cohen's d
TracVLA	Object	d=1.212	empty	Cohen's d
TracVLA	Long	d=1.473	empty	Cohen's d
NORA	Goal	cos=1.0	sat.	cosine

Table 22: Three-tier routing gradient.

Tier	Example	R^2 Coll.	CF	Routing
VLM base	Phi-3-V	None	cos<0.2	Lang-only
VLM base	Qwen-VL	None	0.081	Lang-only
OXE VLA	openvla	None	d=1.05	General
Task VLA	TracVLA	Δ=0.23	d=1.52	Task-spec.
Task VLA	NORA	Δ=0.69	cos=1.0	Task-spec.
NLP ctrl	Llama-chat	Δ=0.00	N/A	None

VLA—reframes VLA training not as monolithic adaptation but as staged routing emergence. The first transition (VLM→VLA) triggers a one-time representational reorganization (exit collapse) and establishes general instruction routing. The second transition (general→task-specific) sharpens counterfactual sensitivity without additional exit collapse. This staged view has practical implications: monitoring exit collapse during training could serve as a diagnostic for whether the fundamental VLM→VLA reorganization has occurred.

What training creates vs. what it inherits. A key distinction is between training-induced effects (R^2 exit collapse, counterfactual sensitivity—consistent across architectures) and base-model-dependent properties (L1 head count—determined by VLM/LLM pretraining). Conflating these would misattribute inherited attention patterns to VLA training.

Metric dependence. The RSA–CKA ordering reversal (Section 4.5) cautions against single-metric conclusions. Our observation that DINOv2 achieves the highest geometric alignment holds under RSA but not CKA, reflecting the distinct representational properties each metric captures. The inverted-U RSA profile at mid-layers is corroborated by functional R^2 probes (metric-independent), but the rise-phase magnitude ($\Delta\rho$ = +0.016) is small and should not be over-interpreted.

Limitations. (1) All analyses are open-loop (no closed-loop robot evaluation); routing changes may not directly translate to behavioral differences. (2) Counterfactual metric incomparability between trained and untrained conditions (5/8 cross-architecture cells) limits

Table 23: L1 instruction-dominant heads: training vs. inheritance.

Arch	Suite	Trained	Untrained	Δ	Source
OpenVLA	Goal	23/32	26/32	−3	Base+OXE
OpenVLA	Object	24/32	24/32	0	Base+OXE
TracVLA	Goal	27/32	32/32	−5	Base VLM
TracVLA	Object	30/32	32/32	−2	Base VLM
NORA	Goal	13/16	13/16	0	Base VLM
NORA	Spatial	12/16	14/16	−2	Base VLM

Table 24: Cross-benchmark consistency (OpenVLA).

Metric	LIBERO-Goal	LIBERO-Spatial	Consistent?
RSA peak (T img)	L12 (0.628)	L14 (0.691)	Yes (+2L)
R^2 peak (T img)	L8 (0.655)	L13 (0.696)	Yes
Untrained RSA peak	L0 (0.612)	L0 (0.652)	Yes
Training Δ	+0.017	+0.023	Both sig.
Untrained collapse	L10–L13	L5–L31	Both

quantitative comparison. (3) The $\Delta > 0.10$ threshold for exit collapse, while empirically validated by the clean 6/6 vs. 0/4 separation, is not derived from a theoretical principle. (4) Our analysis covers three architectures; additional VLA families would strengthen generalization claims.

8 Conclusion

We present a systematic experimental framework for token-level analysis of Vision-Language-Action models, validated across three architectures (OpenVLA, TracVLA, NORA) and four LIBERO task suites. Our analysis reveals that VLA fine-tuning achieves immediate dedicated instruction routing by reorganizing attention from mid-layer processing to entry-exit bookend patterns, that R^2 exit collapse is a one-time signature of the VLM-to-VLA transition (6/6 VLM→VLA cells vs. 0/4 VLA→VLA cells), and that vision encoder geometry is invariant to both training state and instruction content. These findings are unified by a three-tier routing gradient that characterizes instruction routing as a staged emergent property of VLA training.

Table 25: TracVLA mixed heads across suites.

Suite	T Instr-Dom	T Mixed	U Instr-Dom	h13/h22
Goal	27/32	5	32/32	mixed
Spatial	30/32	2	32/32	mixed
Object	30/32	2	32/32	mixed
Long	30/32	2	32/32	mixed

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